A City with Doors

Jing-Zhang | Human-Agent Coexistent Urban Design

REVIEW NOTES

- Keep human life continuous and let personal AI travel with the person across the city. Public facilities are called only by task and permission, and every call remains maintainable and revocable.
- Zhongzhiyuan: R&D, testing and collaboration.

SYSTEM / SPATIAL LOGIC

- AI Origin Community: home, school and community life.
- Dazhongsi: arrival, exchange and continued cooperation.
- Jing-Zhang acts as the public spine. Personal AI stays with the person; R6 robots and Docks are public facilities borrowed for bounded tasks.

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Real City and Three Scales

The existing city is not a blank site.

REVIEW NOTES

- 43.6 km^2: strategic relationships and citywide service resilience.
- 11.4 km^2: overall design of the public spine, nodes and continuous urban support.
- Three key areas: Dazhongsi -> AI Origin Community -> Zhongzhiyuan.
- Six urban breaks are addressed: weak continuity between key areas; unstable school-hospital-home approaches; rail arrival disconnected from street services; incomplete robotics backstage; heat/rain/snow interrupting ordinary routes; and loss of a minimum operating layer during outages.

SYSTEM / SPATIAL LOGIC

- Responses include the Jing-Zhang public spine, safe waiting and crossings, station-city lobbies, R6 logistics/repair, continuous shade and covered routes, low-power communication and explicit human takeover points.
- Scope is based on published textual extents and approximate areas. The working diagram is not a statutory redline, parcel, ownership or engineering basis.

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11.4 km² Overall Design

One public spine weaves ordinary neighbourhoods and the three key areas into one city.

REVIEW NOTES

- North: research, campuses and housing coexist; testing and maintenance backstage do not cut through everyday neighbourhoods.
- Middle: education, healthcare, family life and public space form an all-day living chain around the AI Origin Community.
- South: rail arrival, hotels, meetings, consumption and residential quiet routes are layered around Dazhongsi.

SYSTEM / SPATIAL LOGIC

- Cross-area baseline: Jing-Zhang, ordinary streets, public services, energy and communications carry the continuous relay.
- Shared operating grammar: ordinary passage first; personal-AI task continuity; blue-green public space; energy/communication recovery.
- This page retains the v0.8 overall-map logic as a working baseline and is replaced from the same source when formal spatial data arrives.

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Urban Renewal and Functional Organisation

Preserve city life, then repair broken links as new spatial interfaces.

REVIEW NOTES

- Preserve mature housing, daily-life foundations, campuses and existing public services.
- Repair accessibility, crossings, canopies and the continuous school-community-pharmacy route.
- Reversibly embed visitor lobbies, public-robot points, temporary activities and testing services.
- Keep station-city/business, public services, housing/courtyards, R&D/experiments and repair/charging backstage as distinct spatial roles.
- Zhongzhiyuan exposes R&D and testing to public observation while logistics, repair, charging and safe return-to-base remain in a complete backstage.

SYSTEM / SPATIAL LOGIC

- AI Origin keeps school, community health, housing and daily parks walkably continuous while every household retains an independent front door.
- Dazhongsi grades down from arrival/hotel/meeting/consumption toward residential quiet zones. Jing-Zhang keeps ordinary passage open while task-based facilities open and close as needed.
- Priority: 1) ordinary passage and public service; 2) AI/robot interfaces; 3) large construction dependent on verified formal data.

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Jing-Zhang Public Spine

How personal AI calls urban interfaces and keeps people moving through faults.

REVIEW NOTES

• Ordinary passage stays open. Personal AI runs on the user's own device, reads corridor conditions, requests bounded TASK permissions, calls sites or public robots, receives receipts and revokes access.

• Three corridor segments: Zhongzhiyuan - research and nature; AI Origin - life and learning; Dazhongsi - arrival and exchange.

• Distributed public-service nodes publish state, verify task object/route/area/time, return execution receipts, and keep no personal profile or root access.

SYSTEM / SPATIAL LOGIC

• Non-AI baseline: walking, cycling/accessibility, physical wayfinding, waiting, drinking water, shelter and human help remain usable.

• Typical nodes include the blue-green trial loop, outdoor classroom, station public hall, machine-weather/R6 dock, older-adult rest/home-maintenance node, and reversible activity/quiet-service area.

• Three operating tests: rainy-night reroute; snow/heat opening and maintenance; outage with screen-free help. Restoration resumes from verified checkpoints rather than reconstructing a person's history.

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Spatial and Climate Section

Doors, courtyards, covered walks, trees, water and stone make one sustainably usable urban path.

REVIEW NOTES

- Five spatial types: independent home door; shaded/ventilated courtyard; school/community-service threshold; Dock/elevator/storage/service threshold; Jing-Zhang public recovery space.
- Four conditions cut across all five spaces: heat shade, ventilated resting, storm-water drainage and low-power snow-night operation.

SYSTEM / SPATIAL LOGIC

- Line grammar: grey solid = non-AI passage; dark green = AI task chain; blue dashed = state rerouting; threshold = on-demand authorisation.
- Personal AI carries task, permission and checkpoint continuity across spaces. Doors, Docks and R6 open only for the current task.
- Failure fallback keeps lighting, wayfinding and help available even when higher-level automation is degraded.

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Beijing AI Origin Community

A district where a world-class innovation ecosystem and ordinary daily life coexist.

REVIEW NOTES

- Higher education, research, work/startups, schools, healthcare and nearby family housing form a working 5-10 minute living network around the Jing-Zhang public spine.
- Research/startup route: university interface -> shared lab -> innovation commons -> talent services. Personal AI continues tasks but does not enter residential quiet zones at night.
- Family/education route: independent home -> school -> outdoor classroom -> Jing-Zhang. Pickup changes reroute at safe waiting points; schools receive only necessary receipts.

SYSTEM / SPATIAL LOGIC

- Healthcare/older-adult route: housing -> community health station -> pharmacy -> ordinary transport. Personal AI connects preparation, pickup and return without exposing the full household context.
- Public-service lobby offers ordinary enquiries, a non-AI route, time-limited task applications and human relay. Public interfaces open by task only.
- Nearby generations may live within 5-10 minutes but each household keeps its own front door. The district supports connection without merging households.

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One Ordinary Workday for Three Generations

Personal AI continues learning, healthcare and the trip home without becoming a central household brain.

REVIEW NOTES

• Three personal AIs remain continuous for the child, mother and grandmother. Household shared intelligence coordinates only the temporary common task.

• 07:20 split: the child loads course/safety information; the mother registers her available time; the grandmother creates a one-off medical task pack.

• 09:30 normal operation: school receives course/receipt data; the mother's work schedule continues; the grandmother uses calling, navigation, queueing and payment without merging private memory.

• 15:30 plan change: the mother submits “temporarily unable to pick up”; the grandmother updates her return ETA; the family task layer keeps only pickup time, availability, remote ETA and the completed receipt.

SYSTEM / SPATIAL LOGIC

• 17:05-17:40 recovery: result = “17:05 grandmother receives Xiahe at N2”. The Jing-Zhang N2 node provides low-power wayfinding, safe waiting, drinking water and backup power.

• 20:30 closeout: learning material returns to the child's AI, work schedule to the mother's AI, medical/medication information to the grandmother's AI, then the shared task closes and permissions expire.

• Spatial landings: home local backup; school/hospital task interfaces; named pickup-responsibility seat; N2 recovery node; return-home closeout. Public interfaces return receipts and never read the whole household.

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Near, but Not Cohabiting

Chinese family kinship is organised through courtyards, independent front doors and shared boundaries.

REVIEW NOTES

- Courtyard-scale cluster: independent homes for older and younger generations face a shared courtyard, while each household keeps its own door, kitchen and privacy threshold.
- Robots have explicit tasks, service paths and parking/return locations. Family robots do not wander through another household simply because the families are related.
- Within one home, the entrance, kitchen, bedroom and balcony/greenhouse keep different service and privacy bands; Dock/service areas are separated from intimate rooms.

SYSTEM / SPATIAL LOGIC

- Private intelligence is layered: multiple personal AIs remain separate from one household shared coordinator. The household coordinator handles “who/when/task”, not merged personal memory.
- Public services remain outside the household boundary and receive only task results/receipts.
- Fault decoupling: grid -> community microgrid -> building emergency power -> household low-voltage backup -> lighting/help/mechanical access. Smart degradation must not mean “cannot go home”.

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Dazhongsi Station-City Front Hall

Personal AI strings visitor services together, while lively uses never cross the residents' quiet line.

REVIEW NOTES

- One personal-AI task chain overlays four categories of physical routes. The space becomes quieter and more private the closer it gets to housing.
- Arrival/short-stay line: station arrival, baggage service and visitor guidance.
- Reversible activity line: temporary markets, events and installation testing.
- AI-native service/commerce: meetings, dining, work continuation and product experiences; hotel public lobby acts as a shared interface rather than a private-data sink.

SYSTEM / SPATIAL LOGIC

- Jing-Zhang public life remains ordinary and open. Resident quiet routes and neighbourhood commerce are protected from visitor through-traffic.
- Service backstage handles staging, storage, return, repair and Dock maintenance without crossing the resident quiet line.
- Four must-build nodes: station Dock; hotel public lobby; quiet service line with ordinary enquiry/human support; service backstage for R6/Dock staging and maintenance.
- Visitor-service gateway runs as a bounded interface loaded onto the person's own AI, with receipts and revocation at task end.

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Visiting Scholar Mira - 36 Hours

The same AI continues from invitation to ongoing collaboration.

REVIEW NOTES

• Two equal entry modes: if Mira already has personal AI, load the temporary city interface; if not, issue a continuous visitor-AI service pack for this visit.

• Before travel: invitation becomes a task list covering transport, hotel, dining, language and goals.

• D1 18:00 arrival: delay triggers automatic rescheduling of check-in, evening plan and public-lobby arrival.

• D1/D2 08:30-11:40: meeting-place change causes one coordinated update rather than repeated manual re-entry.

SYSTEM / SPATIAL LOGIC

• D2 12:20: Mira authorises one bounded R6/public-body task to collect items, receive a package and return to Dock with a signed receipt.

• D2->D3: visitor permission closes at departure. Continued experimental collaboration is a new permission explicitly authorised by Mira.

• City endpoints provide only necessary interfaces and results; they do not store the visitor's complete personal-AI context. Ordinary station, lobby, quiet-route and human-service options remain usable if AI is unavailable.

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Zhongzhiyuan AI Autonomous Innovation Acceleration Zone

Garden-based full-stack innovation connects R&D, compute, validation, industrialisation and governance.

REVIEW NOTES

• The garden public spine is an open collaboration front stage for walking, public observation, results display and visitor exchange.

• Full-stack nodes: basic research/collaboration; product engineering/prototyping; compute/data collaboration; machine-weather and integrated validation; standards/governance/safety audit; industrialisation/small-batch production; repair/charging/retest; city trial handoff.

• Three spatial bands: public knowledge front stage; controlled experimental courtyards; maintenance/logistics backstage.

SYSTEM / SPATIAL LOGIC

• Personal AI keeps the innovation task continuous while each node receives only segmented permissions and returns a result receipt.

• Task sequence: 01 research collaboration -> 02 compute/data -> 03 product validation -> 04 industrialisation -> 05 governance feedback and re-entry.

• The diagram is a working relationship model, not an official parcel plan; boundaries, road redlines, ownership, existing buildings and precise locations remain to be verified.

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Zhongzhiyuan R&D Validation Day

A young engineer's joint-experiment day under changing resources, weather and city-trial conditions.

REVIEW NOTES

- 08:30 shared lab: A-Cen and personal AI verify compute, laboratory, device status and visitor permissions.
- 10:15 machine-weather field: engineer, AI and R6 jointly tune equipment and confirm weather thresholds and stop conditions.
- 13:20 compute/microgrid node: grid fluctuation triggers checkpoint persistence and backup power; training is delayed and R6 stops safely.
- 16:00 city-trial handoff: an external team receives a limited task pack and temporary permissions for a bounded cross-area test.

SYSTEM / SPATIAL LOGIC

- 19:00 repair/retest/governance feedback: faults enter repair, retest and re-entry; results update the next validation cycle.
- Operating loop: trigger event -> personal AI evaluates state/risk/priority -> resource rerouting -> spatial response -> results feed back into the next round.
- The page tests spatial and operational interfaces among R&D, compute, electricity, equipment and cross-area trials.

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Public Bodies and Facility Calling

Personal AI stays continuous; public bodies perform only the task in front of them.

REVIEW NOTES

- The city makes “borrow a body, an elevator or a locker” visible as stations, paths, lobbies and return systems.
- A time-limited task credential names object, area, route, validity and the human responsible for exception takeover.
- At task end: return -> clean -> recharge -> clear task cache -> revoke permission.
- Public bodies never carry full personal memory, inherit long-term access or cross home/hotel private domains.

SYSTEM / SPATIAL LOGIC

- Zhongzhiyuan: engineer calls testing equipment/robots between lab, test field and repair harbour.
- AI Origin: resident calls an assistive robot from community/home to metro/service point, then returns it after the errand.
- Dazhongsi: visitor calls baggage or guidance robots between arrival, transfer Dock and hotel lobby.
- One shared closed loop: AI initiates -> bounded credential -> public body -> receipt -> return/revocation.

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Energy, Communications and Recovery

First keep people able to move and ask for help; then preserve critical services.

REVIEW NOTES

- Layered resilience combines grid power, district backup, building/public-node UPS, low-power communications, edge checkpoints and robot safe-return Docks.
- AI Origin: backup for community public services/healthcare, key building circuits and care; non-essential home automation may stop.
- Dazhongsi: station-city services/wayfinding, transport/hotel credentials and emergency charging continue; advertising/event screens may stop.
- Zhongzhiyuan: microgrid/experimental buildings, critical equipment, machine-room UPS and robot safe stop continue; non-critical high-energy training may stop.

SYSTEM / SPATIAL LOGIC

- Jing-Zhang: N2/line-side service points, low-power wayfinding/broadcast/help, communication relay and safe stopping continue; decorative interactive installations may stop.
- 21:40 outage sequence: immediate safety switch -> short-term local degradation -> long-duration critical relay -> restoration verification -> item-by-item restart from verified checkpoints.
- Three redundancies: route redundancy (walking/covered route/stairs/manual opening), facility redundancy (manual bypass/inspection/isolated zones), information redundancy (phone/paper/physical signs/broadcast/fixed human window).

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Personal-AI Continuity and Machine Lifecycle

People can change devices or services without rebuilding every collaboration from zero.

REVIEW NOTES

- Personal AI continues the resident's work; backups stay under resident control; hardware is independently repairable; public systems receive only bounded task credentials.
- Five stages: carry continuity -> enter a home with doors -> borrow a public carrier -> switch carriers after failure -> continue on the new carrier.
- Bounded credential fields: task, destination, callable facilities, validity and takeover responsibility. Completion returns a receipt and revokes permission.
- The full long-term memory, family-relationship data and root permissions of personal AI stay inside the private domain.

SYSTEM / SPATIAL LOGIC

- Continuity capsule: identity anchor/recovery index, emergency contacts, care rules/key schedule and recent necessary state.
- Three backups: near-term recovery copy on current/private devices; isolated periodic snapshot in a resident-designated independent store; resident-held migration package on encrypted local media/personal data store.
- Public systems may help verify, recover and transfer but never hold the only recovery key or a full private copy.
- Hardware service chain: inspection/diagnosis -> repair/replace -> retest -> re-entry. Dazhongsi supports discovery/rental, AI Origin daily use, local merchants repair/substitute devices, and Zhongzhiyuan complex retest/re-entry.

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Projects, Phasing and Operation

Ten projects sit on one city baseline and open gradually under explicit urban responsibility.

REVIEW NOTES

- P01 blue-green open validation loop; P02 machine-weather and safe return; P03 repair/charging and edge backstage; P04 15:30 three slow doors; P05 visible help-door network; P06 21:40 continuity node; P07 station-city four-quadrant stitching; P08 public lobby and short-visit interface; P09 activity commons and resident quiet spine; P10 cross-area service station.
- Phasing: 0 preserve the ordinary baseline; 1 embed reversible interfaces; 2 conduct bounded testing/retesting; 3 expand or stop according to evidence.
- Responsibility chain covers use, construction, operation, maintenance, takeover and stopping - responsibility is not outsourced to “the system”.

SYSTEM / SPATIAL LOGIC

- Year-round operation does not mean year-round events: spring classroom/community planting; summer shade/night walking/device cooling; autumn developer exchange/public review; winter snow drills/repair.
- Reference mechanisms include Modena bounded test areas, Detroit multimodal service access, SHOW joint evaluation, Punggol phased fixed stations, Yizhuang permission ladders and Waymo curb/maintenance/remote support.
- Points, responsible entities, investment, capacity and timing still require verification against formal redlines, ownership and engineering conditions.

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Scenarios, Evidence and Closing

Twelve checks return to the same city and test whether personal-AI tasks, permissions and checkpoints can continue, pause and recover.

REVIEW NOTES

- Three long life lines: Zhongzhiyuan engineer - resource/test/repair/retest; AI Origin three-generation family - learning/healthcare/change/return home; Dazhongsi visiting scholar - arrival/borrowing/collaboration/return.
- Nine task nodes: Z1-Z3, A1-A3, D1-D3. Three Jing-Zhang fault checks: J1 rainy-night reroute, J2 maintenance takeover/permission reroute, J3 outage help/offline continuation.
- Evidence is reused across five endpoints: A3 booklet, A0 board, offline HTML, GeoJSON/metrics and the formal report.
- Coverage ledger: 3 long scenarios; 12 validation nodes; P01-P10; 4 conceptual cultural nodes; 3 Zhongzhiyuan validation types; 6 mechanism references; at least 5 role types.

SYSTEM / SPATIAL LOGIC

- Three spatial indicators stay source-consistent: 11.4 km^2 announced overall-design area; green-space ratio recalculated from the same geometry; public-space ratio recalculated from the same geometry. No placeholder value is invented.
- Evidence states distinguish public, working, designed and pending-verification material.
- When official geometry arrives, scope/land use/roads/green/buildings/phasing -> metrics -> all five endpoints are replaced and recalculated together.
- Closing rule: personal AI retains task, permission and checkpoint continuity; Jing-Zhang interfaces provide fault rerouting. Resume only after verification; close permissions when the task ends.

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